

Applying Machine Learning to Communication Churn Prediction in Multichannel CRM Platforms: Email, Push, and In-App

Abraham Trenyan

Abstract: Multichannel CRM platforms detect user disengagement reactively, once the loss of interaction has already occurred. This work proposes a supervised classification model that combines behavioral signals from email, push, and in-app channels to predict communication churn before it materializes. Communication churn is defined as the complete cessation of interaction with messaging channels while the user remains active on the platform, a distinction not explored in the existing literature. Using public data from Kaggle, Python, and the Random Forest and XGBoost algorithms, a prediction pipeline is built, evaluated with AUC-ROC and recall, and designed to integrate directly into a CRM platform and trigger re-engagement campaigns automatically. The proposal outlines a viable path to transform the detection of communication disengagement into a predictive and actionable process within modern CRM platforms.

I INTRODUCTION

Users of digital platforms are exposed to a very high volume of brand communications through channels such as email, push notifications, and in-app messages. Empirical studies conducted in mobile contexts report medians of 56 to 63 notifications per user per day [1], a stimulus load that has been associated with negative emotions and with the dismissal or uninstallation of the applications responsible [1, 2]. In the specific domain of email marketing, the analysis by Chaparro-Peláez and colleagues of more than five thousand campaigns sent to hundreds of millions of users identifies a significantly negative relationship between sending frequency and open rate [3], and Zhang, Kumar, and Cosguner show that sending above or below the optimal frequency can reduce the expected profitability per customer over their lifetime by sixteen to thirty-two percent [4]. Taken together, this evidence suggests that engagement with direct communication channels is sensitive to communication intensity, and that disengagement can manifest progres-

sively before a total abandonment.

Communication disengagement also has a direct economic impact. Classic work on retention estimates that a five percent reduction in the defection rate can generate profit increases of between twenty-five and eighty-five percent depending on the industry [5], and that a one percent improvement in retention increases firm value by approximately five percent [6]. In the case of mobile communications, users who receive push notifications during their first ninety days exhibit retention rates approximately three times higher than those who receive none [7].

CRM platforms manage their user base through segmentation strategies: users are grouped by common characteristics and receive campaigns designed for that profile [8, 9]. This approach has been the standard for years, but segments are defined with fixed rules that do not adapt in real time to each user's individual behavior: by the time a user is classified as inactive or at risk, in many cases they have already gone weeks without interacting with any channel. Research on multichannel CRM environments confirms that this reactive model leaves considerable room for improvement, especially when behavioral data exist but are not exploited in an integrated way [8, 9, 10].

The field of machine learning offers tools that can change this dynamic. Supervised classification models, supported by solid theoretical foundations [11], learn patterns from historical data and can anticipate events before they occur. Algorithms such as Random Forest [12] and XGBoost [13] have demonstrated outstanding performance in classification tasks on complex tabular data, including churn prediction in banking [14], telecommunications [15, 10], and digital services [16]. Their wide adoption in applied research, their availability as open-source tools, and their ability to handle heterogeneous tabular data in class-imbalanced scenarios make them especially appropriate candidates for the problem at hand [13, 11, 14, 16]. The evaluation of these models typically relies on metrics such as AUC-ROC [17], which allow classifiers to be compared independently of the selected decision thresh-

old. Beyond product churn, that is, the cancellation of an account, machine learning applied to user behavior analysis is a mature methodological line, with consolidated applications in areas such as advertising response prediction and the analysis of digital interactions [18, 19]. Systematic reviews specific to the churn prediction field confirm important advances in ensemble algorithms and deep learning architectures, but also identify persistent challenges related to class imbalance, model interpretability, and comparability across studies [20, 15]. These reviews concentrate mostly on product churn in domains such as telecommunications, banking, or subscription services, leaving the user’s communication behavior across multiple messaging channels in the background.

This gap becomes especially visible when considering what can be called communication churn: the moment when a user stops interacting with all of a brand’s messaging channels while remaining active on the platform. Unlike product churn, communication churn is a subtler and earlier phenomenon. A user who stops opening emails, ignores push notifications, and does not respond to in-app messages has not abandoned the platform, but has become effectively unreachable. From the perspective of customer engagement theory, which describes consumer involvement in terms of cognitive, emotional, and behavioral dimensions [21, 22], this represents a critical inflection point at which preventive intervention is still possible.

This work proposes to address that gap by building a supervised classification model that combines behavioral signals from the three channels, email, push, and in-app, to predict the risk of communication churn before it occurs. The model is built with Python using the scikit-learn and XGBoost libraries, trained on public data available on Kaggle. Model performance is evaluated with AUC-ROC and recall, metrics suited to scenarios where the prevalence of the positive class must be determined empirically and where the cost of missing a positive case exceeds that of an incorrect activation. The final goal is for the resulting risk score to be integrated directly into a CRM platform and to trigger re-engagement campaigns automatically, transforming disengagement detection from a reactive process into a predictive one.

The body of this work is organized as follows. Section II reviews the state of the art in churn prediction, CRM segmentation, user behavior analysis, and communication fatigue. Section III presents the theoretical framework, covering the foundations of su-

pervised learning, the selected classification models, and the evaluation metrics. Section IV describes the methodology, including data preparation, feature engineering, and model training. Section V presents the results and analysis. Section VI offers conclusions and possible lines of future work.

II State of the art

A Churn prediction with machine learning

Customer churn prediction is one of the best-studied problems in machine learning applied to marketing and services. Recent literature concentrates on domains where account cancellation is an explicit and measurable event, mainly telecommunications [10, 15], retail banking [14], and subscription-based digital services [16]. Tree-based ensemble algorithms have become the methodological reference, largely due to their performance on tabular data and the availability of efficient implementations in open-source libraries [14, 16].

Systematic reviews of the field confirm that churn prediction has progressively incorporated more sophisticated techniques, from classical models toward ensembles and deep learning architectures, while simultaneously identifying persistent challenges linked to class imbalance, model interpretability, and the lack of comparability across studies due to the heterogeneity of datasets and metrics [15, 20]. These reviews, however, keep the focus on churn understood as account cancellation, without addressing earlier or partial forms of user disengagement.

B Segmentation and multichannel CRM

Customer relationship management in multichannel platforms traditionally relies on segmentation strategies that group users according to demographic, transactional, or behavioral characteristics, and that assign campaigns designed for each segment [8]. The literature on segmentation methods in e-commerce shows an evolution from heuristic rules toward data-driven clustering techniques, but largely preserves a static logic, in which segments are defined periodically and are not updated at the pace of individual behavior [8].

The review by Verhoef and colleagues on CRM in multichannel retail environments argued, more than a decade ago, that the potential of data-rich platforms is underutilized when interaction data exist but are not effectively integrated into operational decision models [9]. That observation remains valid: most current

CRM implementations maintain a reactive approach, in which intervention arrives after disengagement has already manifested, and rarely incorporate predictive models that combine signals from multiple messaging channels in real time.

C User behavior analysis

User behavior analysis supported by machine learning is a mature methodological line, with consolidated applications in personalization, recommender systems, fraud detection, and response modeling [18]. The review by Martín and colleagues offers a broad map of the field, identifying the most widely used models, the typical data sources, and the types of tasks addressed, and makes clear that digital user behavior has been studied intensively from multiple angles [18].

A particularly developed subfield is response prediction in online advertising, where the central problem consists of estimating the probability that a user will interact with an ad, either through a click or a subsequent conversion [19]. The techniques employed in this domain share a common methodological basis with churn prediction, since both problems are formulated as binary classification tasks on behavioral data with imbalanced classes. However, the emphasis has historically been placed on predicting positive response events, leaving the predictive modeling of sustained absence of interaction less explored.

D Communication frequency and user fatigue

The relationship between the intensity of brand communications and user response has been studied in depth in both the mobile notification and email marketing channels. In the case of push notifications, the in-situ study by Pielot and colleagues quantifies the volume to which users are exposed and associates that volume with negative emotions and with the decision to dismiss or uninstall applications [1], while Okoshi and colleagues address the cognitive load generated by interruptive notifications and explore mechanisms to mitigate their impact [2]. On the positive side, the technical report by Airship documents significant differences in retention rates between users who receive push notifications during their first days of use and those who do not [7].

In the email marketing domain, the large-scale analysis by Chaparro-Peláez and colleagues identifies a negative relationship between sending frequency and email open rates [3], and the dynamic model by Zhang, Kumar, and Cosguner shows that deviating from the

optimal frequency, whether by excess or by defect, deteriorates the expected profitability per customer over their lifetime [4]. This evidence is framed within a consolidated field of research, but one generally addressed channel by channel in isolation, without integrating email, push, and in-app signals into a single model of user behavior.

III Theoretical framework

A Supervised machine learning

Supervised machine learning consists of inducing a prediction function from a set of examples in which each observation is accompanied by a known label [11]. Unlike unsupervised learning, which searches for latent structures in unlabeled data, supervised learning has a target variable that guides training and allows the prediction error to be quantified directly. When that target variable is categorical and takes two possible values, the problem is called binary classification.

Communication churn, defined as the cessation of interaction with messaging channels while the user remains active on the platform, fits naturally into this paradigm. For each user it is possible to build a vector of behavioral features and assign a binary label indicating whether or not they entered a state of communication churn within a defined time window. The model then learns to map that behavior vector to a churn probability, generating an actionable score for the CRM platform.

B Evaluation and generalization

A model's ability to generalize to unseen data is the fundamental property that determines its practical usefulness [11]. A model that memorizes the training data without capturing genuine patterns, a phenomenon known as overfitting, exhibits deceptively high performance during training and poor performance when applied to new data. To mitigate this risk, the dataset is partitioned into at least two subsets: a training set, on which the model is fitted, and a test set, reserved for estimating performance on new data.

When data are limited or when hyperparameter configurations need to be compared, cross-validation is used, repeating the partitioning procedure several times over different subsets and averaging the resulting metrics. In classification problems with imbalanced classes, such as communication churn, the appropriate variant is stratified cross-validation, which

preserves the proportion of each class in all folds and prevents any fold from being left without positive cases.

C Tree-based models

Decision trees recursively partition the feature space through simple conditional rules, building a hierarchical structure of nodes in which each leaf represents a prediction [11]. In classification tasks, each split is chosen by optimizing an impurity measure, typically the Gini index or entropy, with the goal of separating the classes as cleanly as possible. The main advantage of a tree is interpretability, since its decisions can be followed branch by branch. The main limitation is the tendency to overfit, especially when it is allowed to grow to pure leaves.

The Random Forest method, proposed by Breiman [12], mitigates that limitation by training an ensemble of trees on bootstrap samples of the training set and averaging their predictions. Each tree considers only a random subset of features at each split, which decorrelates the ensemble’s predictions and significantly reduces variance. Random Forest handles heterogeneous tabular data effectively, does not require prior scaling of numerical features, and provides a natural measure of variable importance, properties that make it especially well suited to classification problems with behavioral features such as those in the present work.

D Gradient Boosting and XGBoost

Gradient boosting methods represent an ensemble strategy distinct from that of Random Forest. Instead of training independent trees in parallel and averaging their predictions, boosting builds a sequence of trees in which each new tree specializes in correcting the errors made by the previous partial ensemble. This sequential orientation makes it possible to achieve lower bias than bagging, which trains models in parallel, at the cost of greater sensitivity to noise in the data and the need for explicit regularization to avoid overfitting.

XGBoost, proposed by Chen and Guestrin [13], is a gradient boosting implementation that incorporates regularization in the objective function, native handling of missing values, and algorithmic optimizations that have made it a reference for classification problems on tabular data. For class-imbalanced contexts, XGBoost provides the `scale_pos_weight` parameter, which weights the error on the positive

class during training and compensates for the asymmetry without requiring prior oversampling of the data.

E Logistic regression

Logistic regression models the probability that an observation belongs to the positive class as a sigmoid function of a linear combination of the features [11]. Its output can be interpreted directly as a probability, and the coefficients associated with each feature express the impact of that variable on the log-odds ratio, a property that makes it a reference option when an interpretable model is sought. As a counterpart, it assumes a linear relationship between the transformed features and the logit of the target, a limitation that motivates the complementary use of nonlinear models such as tree-based ones.

F Evaluation metrics for imbalanced classification

The accuracy metric, although intuitive, is misleading in scenarios with imbalanced classes. A trivial model that assigns all cases to the majority class can achieve apparently high accuracy without having learned anything useful. In the communication churn problem the prevalence of the positive class is not known a priori and may turn out to be either minority or majority depending on the dataset and the evaluation window, which requires metrics that reflect the model’s behavior on each class separately rather than on the aggregate. The fundamental metrics in this context are precision, which measures the fraction of predicted positives that are truly positive, and sensitivity or recall, which measures the fraction of actual positives that the model manages to recover.

To evaluate performance independently of the decision threshold, the ROC curve is used, which plots the true positive rate against the false positive rate across all possible thresholds, together with its area under the curve or AUC-ROC [17]. An AUC-ROC value close to one indicates that the model systematically assigns higher scores to positive cases than to negative ones, regardless of the threshold chosen for the final decision. In the present work, recall and AUC-ROC are prioritized as evaluation metrics, given that the operational cost of failing to detect a user entering communication churn exceeds that of generating an unnecessary re-engagement activation.

IV Methodology

A Dataset and case study

The empirical validation of the model is carried out on the public dataset *E-commerce Multichannel Direct Messaging 2021-2023*, published by Michael Kechinov on Kaggle under the Community Data License Agreement. The dataset contains approximately 708 million records of messages sent by a medium-sized e-commerce company over a two-year period, collected through the REES46 CDP platform. Each record represents an individual message sent to a customer and includes the sending channel, the campaign type (bulk, trigger, or transactional), the timestamps of sending and of each subsequent interaction, and binary outcome variables recording open, click, attributed purchase, bounce, spam complaint, block, and unsubscribe events. The dataset comprises 16.8 million unique customer identifiers as message recipients, of which 9.9 million received at least one message during the evaluation window and constitute the evaluable population. It is complemented by three auxiliary files: campaign metadata, first purchase date per customer, and a calendar of commercial events relevant to the analyzed period.

The channels with the largest representation are email and mobile push, which concentrate the entirety of the significant messaging volume in the dataset. The in-app channel, which is part of the theoretical concept of communication churn as introduced in Section I, is not available in public datasets with the event-level granularity required for this work. The methodology is therefore designed over the two highest-volume channels, maintaining an extensible structure that would allow a third messaging channel to be incorporated without altering the processing pipeline or the operational definition of the target.

B Target definition and temporal split

Predicting communication churn requires an operational definition that distinguishes the cessation of interaction with messaging channels from the cessation of activity on the platform. For this work, the target is defined at the user level over a temporal evaluation window: a user receives the communication churn label if during that window they were reached by at least one message on any of the available channels but recorded no open or click on any channel. Users who opened or clicked at least one message during the evaluation window receive the complementary label,

and users who received no messages in that period are excluded from the analysis as they are not informative about the phenomenon under study.

The data partition follows a strict temporal scheme rather than a random one, with the goal of avoiding leakage of future information into the training set. The first eighteen months of the dataset constitute the observation period, over which each user’s behavioral features are built, and the last six months constitute the evaluation period, over which the target is determined. This separation guarantees that the model never observes information from the period in which the variable to be predicted is defined, replicating the deployment conditions of a real CRM environment in which features are computed over the past and the prediction operates on the future.

C Feature engineering

From the observation period, a feature vector is built per user that combines behavioral signals from each individual channel with derived indicators capturing cross-channel dynamics and recent temporal evolution. For each channel, the following are computed: aggregate response metrics (total open and click rates), recent response metrics (open rate over the last thirty and sixty days of the observation period), recency metrics (days elapsed since the last open and since the last click), volume metrics (number of messages received), and friction metrics (counts of hard bounces, soft bounces, unsubscribes, and spam complaints). Additionally, for each user the number of purchases attributed to messaging interaction is computed.

The cross-channel feature block incorporates the number of distinct channels in which the user opened at least one message, the difference between open rates per channel, and the ratio between activity in the last thirty days and the historical average, an indicator designed to detect recent drops in communication engagement. To these features are added two descriptors of campaign context, derived from the join with campaign metadata: the proportion of trigger-type messages over the total and the proportion of messages with personalization elements. Finally, customer tenure is incorporated, measured as the number of days elapsed from the first recorded purchase to the close of the observation period. The resulting feature set is persisted to disk to avoid recomputation in successive iterations of the experiment.

D Models, imbalance handling, and evaluation protocol

Three classification models are trained: a logistic regression serving as an interpretable baseline, and a Random Forest and an XGBoost as the main tree-ensemble models. The selection follows the conceptual coverage developed in the theoretical framework and the comparability with the churn prediction literature, which consistently reports competitive performance for ensemble models on tabular data [14, 16]. Numerical features are standardized only for the logistic regression case, while the tree-based models are trained on the original features, since they do not require prior scaling.

Imbalance handling is applied conditionally on the observed proportion of the positive class. If the frequency of communication churn in the training set is below twenty percent, synthetic oversampling via SMOTE [23] is applied exclusively to the training set, preserving the original distribution in the test set so that metrics reflect the real prevalence of the phenomenon. For Random Forest and XGBoost, a hyperparameter search is performed over a bounded grid, optimizing AUC-ROC through stratified five-fold cross-validation. The logistic regression is trained with default regularization and no additional search, in its role as baseline.

Each model is evaluated on the test set, defined by the temporal partition described previously, reporting five metrics: AUC-ROC, precision, recall, F1-score, and the confusion matrix associated with the default decision threshold. Additionally, for Random Forest and XGBoost, feature importance plots are built to interpret which behavioral signals are most relevant in predicting communication churn. The three models are compared in a summary table synthesizing performance on the aforementioned metrics.

V ANALYSIS AND RESULTS

A Experimental setup

Applying the pipeline described in Section IV to the full dataset identified 9,906,011 users with at least one message received during the evaluation window. A stratified eighty-twenty split was applied to that population, preserving the target prevalence in both subsets. The resulting partition contains 7,924,808 training users and 1,981,203 test users.

The observed prevalence of the positive class reached fifty-seven percent in both subsets. This distribution

indicates that, among the population reached by messaging during the evaluation window, the majority of users recorded no open or click on any channel. The operational criterion defined in Section IV.D established the activation of synthetic oversampling via SMOTE [23] only when the prevalence of the positive class was below twenty percent. Since the condition was not met, the pipeline trained the three models on the original distribution without applying oversampling.

The hyperparameter search for Random Forest and XGBoost was executed through stratified five-fold cross-validation with twenty random sampling iterations over the hyperparameter space, optimizing AUC-ROC. The search operated on a stratified subset of 500,000 users drawn from the training set, and the winning configuration of each model was then refitted on the full training set so that the final model leveraged all available data. The logistic regression was trained directly on the full set with default regularization, in its role as interpretable baseline.

B Model comparison

Each model was evaluated on the test set at the decision threshold that maximizes F1. To quantify the uncertainty of the metrics, bootstrap with two hundred iterations was applied to the test set, reporting ninety-five percent confidence intervals by the percentile method. Table 1 summarizes the results.

Random Forest achieved the best AUC-ROC performance, with an improvement of 0.036 points over the logistic regression, and stands as the winning model under the evaluation criterion defined in Section IV.D. XGBoost obtained metrics statistically indistinguishable from Random Forest, with an AUC-ROC difference on the order of one thousandth, within the variability range estimated by bootstrap. The gap over the logistic regression confirms the pattern consistently reported in the churn literature on tabular data, in which tree ensembles outperform linear models by margins on the order of tenths in AUC-ROC [14, 16].

The recall of all three models sits above 0.94. This behavior is consistent with the majority character of the positive class in the evaluated population: the F1-maximizing threshold biases predictions toward the predominant class, recovering most of the positives at the cost of a moderate precision close to 0.70. The asymmetry is consistent with the focus of this work, in which the operational cost of failing to detect a user in a state of communication churn exceeds the cost of generating an unnecessary re-engagement activation.

The compared ROC and Precision-Recall curves

Table 1: Model comparison on the test set (1,981,203 users) at the F1-maximizing threshold. Ninety-five percent confidence intervals by bootstrap (two hundred iterations).

Model	AUC-ROC	PR-AUC	F1	Precision	Recall
Logistic regression	0.771 [0.771, 0.772]	0.773 [0.772, 0.774]	0.791 [0.790, 0.791]	0.675 [0.674, 0.675]	0.954 [0.954, 0.955]
Random Forest	0.807 [0.806, 0.808]	0.821 [0.821, 0.822]	0.804 [0.803, 0.804]	0.701 [0.700, 0.701]	0.943 [0.942, 0.944]
XGBoost	0.806 [0.805, 0.806]	0.820 [0.819, 0.820]	0.803 [0.803, 0.804]	0.698 [0.698, 0.699]	0.946 [0.945, 0.947]

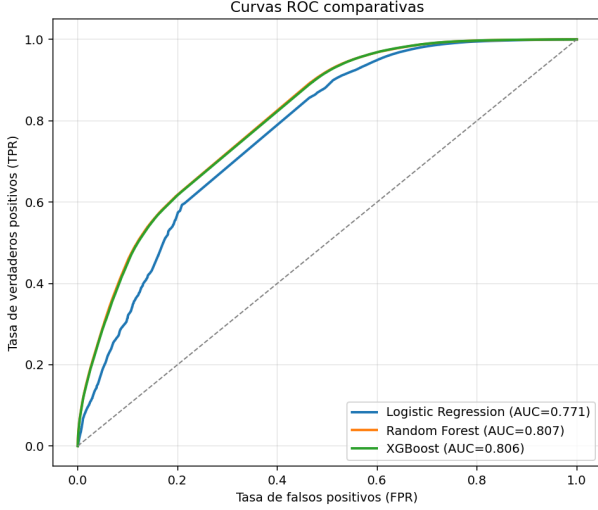


Figure 1: ROC curves on the test set for the three models.

for the three models are shown in Figure 1 and Figure 2 respectively. The confusion matrices associated with each model’s optimal F1 threshold are presented in Figures 3, 4, and 5.

C Feature importance

Figure 6 and Figure 7 show the relative importance of each feature for Random Forest and XGBoost respectively. Random Forest reports importance through the mean impurity reduction accumulated by each variable across all trees in the ensemble, while XGBoost uses each variable’s average contribution to the model’s gain at its splits.

Both models agree in ranking the ratio between activity in the last thirty days and the historical average as the most important feature, by a wide margin in the case of XGBoost, confirming that a recent drop in engagement relative to each user’s own history is the strongest signal of imminent communication churn. The next positions are dominated by recent open counts and open rates over the thirty- and sixty-day windows for both channels, together with the recency of the last email open, indicating that the user’s most recent behavior is the most reliable in-

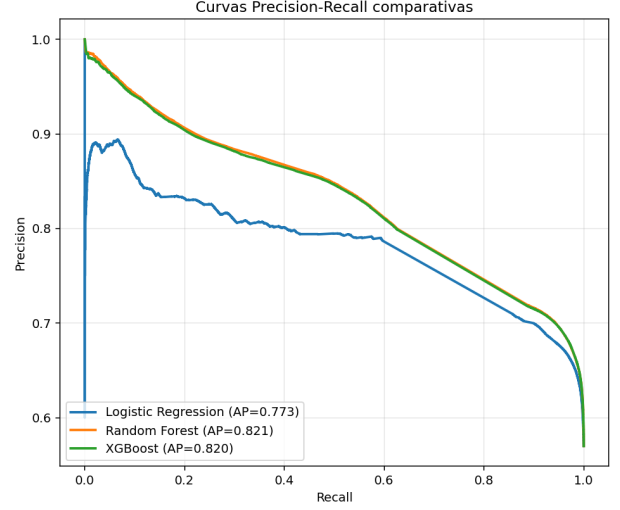


Figure 2: Precision-Recall curves on the test set.

indicator of their current communication engagement state. Customer tenure and total message volume received also appear among the top-ranked features, while friction features, such as hard bounce counts, occupy intermediate positions and campaign context descriptors do not reach the top of either ranking.

D Empirical limitations

Applying the pipeline to real data revealed three limitations worth documenting to properly frame the interpretation of the results.

The first concerns the observed prevalence of the target. On the analyzed dataset the prevalence reached fifty-seven percent, which makes the imbalance problem less pressing but, simultaneously, raises the bar that a predictive model must clear to add value over a trivial strategy of labeling all users as churners. The difference between the AUC-ROC of the best model, 0.807, and that of a random classifier, 0.500, quantifies precisely that contribution.

The second concerns the data quality of the mobile push channel. The features derived from this channel exhibit high proportions of missing values, reflecting that a majority fraction of users either did not receive push messaging or did not record enough

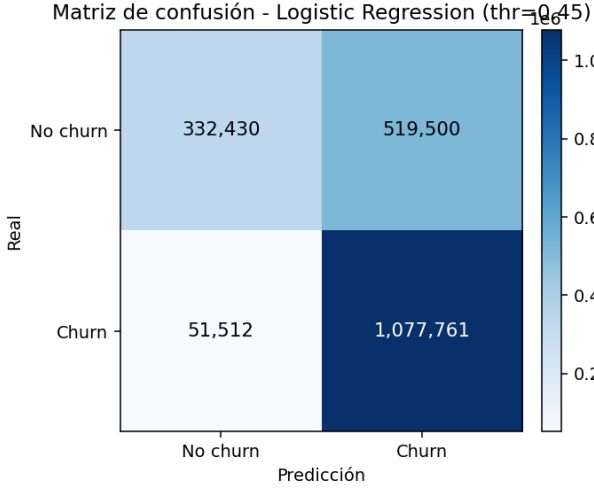


Figure 3: Confusion matrix for logistic regression at the optimal F1 threshold.

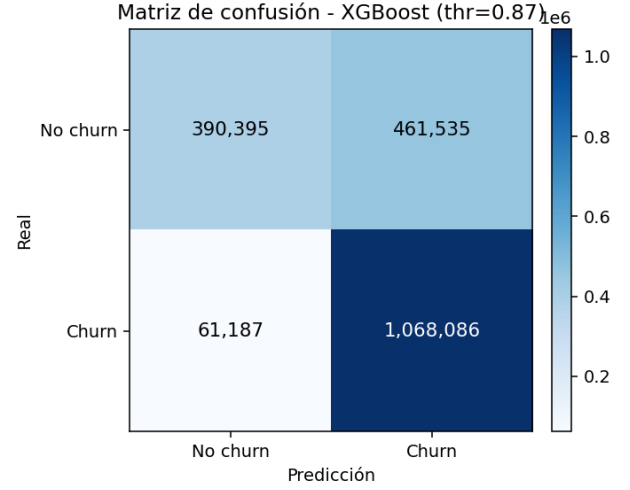


Figure 5: Confusion matrix for XGBoost at the optimal F1 threshold.

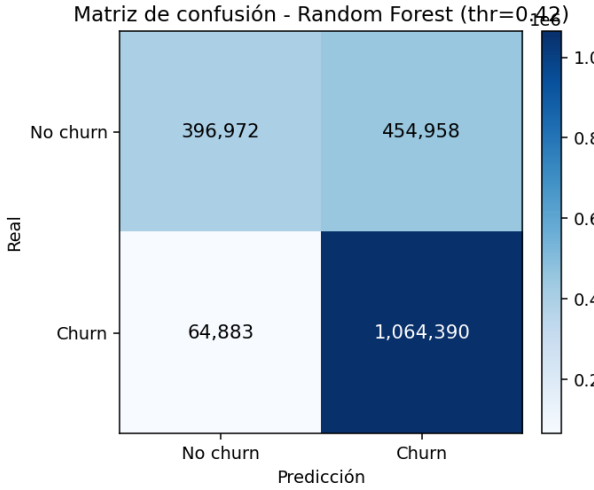


Figure 4: Confusion matrix for Random Forest at the optimal F1 threshold.

interactions for the rate metrics to be computable. Imputation with the median of the observed subset introduces a known bias that should be kept in mind in any causal interpretation of the importance of variables associated with that channel.

The third affects the customer tenure feature. The auxiliary file of first purchase dates covers a period that begins after the cutoff of the observation window, so that a majority fraction of users has no date recorded within the range used and the remaining fraction shows negative numerical values. The feature retains binary predictive power in the sense that distinguishing between users with and without a recorded date is informative, but the numerical tenure value

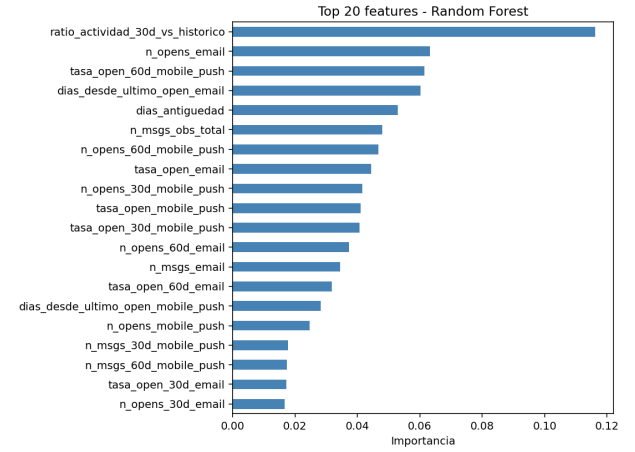


Figure 6: Feature importance for Random Forest.

does not admit the expected causal interpretation. Tree-based models absorb this pattern without requiring additional transformation, but the qualitative reading of this variable in the importance analysis should be nuanced.

The confidence intervals reported in Table 1 are narrow due to the size of the test set, close to two million users. The real variability across model retrainings, not captured by the bootstrap over the test set, stems fundamentally from the hyperparameter search and the internal random sampling of each ensemble algorithm.

VI CONCLUSION

This work proposed a supervised classification model based on ensemble algorithms to predict communi-

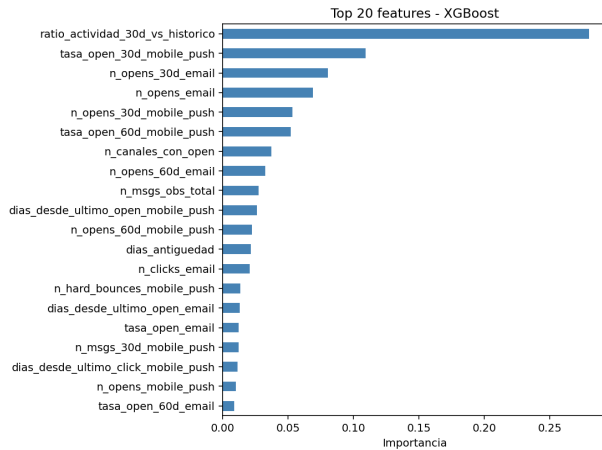


Figure 7: Feature importance for XGBoost.

cation churn in multichannel CRM platforms, a phenomenon defined as the cessation of interaction with messaging channels without account cancellation, and which until now had not been specifically addressed in the academic literature.

The research was successful in that it managed to formalize communication churn as a differentiated object of study, establish a viable methodological framework supported by metrics appropriate for the class imbalance characteristic of the problem, and define a predictive pipeline that can be integrated into operational CRM platforms. Among the limitations encountered, the following stand out: the scarcity of public datasets that combine signals from the three channels in an integrated way, the difficulty of validating the model under real production conditions without a test CRM environment, and the need to balance sensitivity and precision so that the system neither generates fatigue through false positives nor lets real cases of disengagement slip through.

As lines of future work, the following are contemplated: validation of the model in an operational CRM environment, the incorporation of additional temporal signals such as time between interactions and response patterns throughout the day, the exploration of deep learning architectures for sequential data, and the comparative measurement of the predictive model's performance against traditional static segmentation approaches, with the goal of quantifying the operational and economic benefit of replacing fixed rules with dynamic predictions.

VII REFERENCES

- [1] M. Pielot, K. Church, and R. de Oliveira, "An in-situ study of mobile phone notifications," in *Proceedings of the 16th International Conference on Human-Computer Interaction with Mobile Devices and Services (MobileHCI '14)*. ACM, 2014, pp. 233–242. [Online]. Available: <https://dl.acm.org/doi/10.1145/2628363.2628364>
- [2] T. Okoshi, J. Ramos, H. Nozaki, J. Nakazawa, A. K. Dey, and H. Tokuda, "Attelia: Reducing user's cognitive load due to interruptive notifications on smart phones," in *IEEE International Conference on Pervasive Computing and Communications (PerCom)*, 2015, pp. 96–104. [Online]. Available: <https://ieeexplore.ieee.org/document/7146515>
- [3] J. Chaparro-Peláez, Á. Hernández-García, and Á. J. Lorente-Páramo, "May I have your attention, please? an investigation on opening effectiveness in e-mail marketing," *Review of Managerial Science*, vol. 16, no. 7, pp. 2261–2284, 2022. [Online]. Available: <https://link.springer.com/article/10.1007/s11846-022-00517-9>
- [4] X. Zhang, V. Kumar, and K. Cosguner, "Dynamically managing a profitable email marketing program," *Journal of Marketing Research*, vol. 54, no. 6, pp. 851–866, 2017. [Online]. Available: <https://journals.sagepub.com/doi/10.1509/jmr.16.0210>
- [5] F. F. Reichheld and W. E. Sasser, "Zero defections: Quality comes to services," *Harvard Business Review*, vol. 68, no. 5, pp. 105–111, 1990. [Online]. Available: <https://hbr.org/1990/09/zero-defections-quality-comes-to-services>
- [6] S. Gupta, D. R. Lehmann, and J. A. Stuart, "Valuing customers," *Journal of Marketing Research*, vol. 41, no. 1, pp. 7–18, 2004. [Online]. Available: <https://journals.sagepub.com/doi/10.1509/jmkr.41.1.7.25084>
- [7] Airship, "How push notifications impact mobile app retention rates," Airship (Urban Airship), Tech. Rep., 2018. [Online]. Available: <https://grow.urbanairship.com/rs/313-QPJ-195/images/airship-how-push-notifications-impact-mobile-app-retention-rates.pdf>
- [8] M. Alves Gomes and T. Meisen, "A review on customer segmentation methods for personalized customer targeting in e-commerce use cases," *Information Systems and e-Business Management*, vol. 21, 2023. [Online]. Available: <https://link.springer.com/article/10.1007/s10257-023-00640-4>
- [9] P. C. Verhoef, R. Venkatesan, L. McAlister, E. C. Malthouse, M. Krafft, and S. Ganesan, "CRM in data-rich multichannel retailing environments: A review and future research directions," *Journal of Interactive Marketing*, vol. 24, no. 2, 2010. [Online]. Available: https://www.researchgate.net/publication/228711895_CRM_in_Data-Rich_Multichannel_Retailing_Environments_A_Review_and_Future_Research_Directions
- [10] M. G. Abdelhady and K. A. Mohamed, "Leveraging artificial intelligence for predictive customer churn modeling in telecommunications: a framework for enhanced customer relationship management," *Scientific Reports*, vol. 15, 2025. [Online]. Available: <https://www.nature.com/articles/s41598-025-30108-z>
- [11] T. Hastie, R. Tibshirani, and J. Friedman, *The Elements of Statistical Learning: Data Mining, Inference, and*

- Prediction*, 2nd ed. New York: Springer-Verlag, 2009. [Online]. Available: <https://link.springer.com/book/10.1007/978-0-387-84858-7>
- [12] L. Breiman, "Random forests," *Machine Learning*, vol. 45, no. 1, 2001. [Online]. Available: <https://link.springer.com/article/10.1023/A:1010933404324>
- [13] T. Chen and C. Guestrin, "XGBoost: A scalable tree boosting system," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*. ACM, 2016, pp. 785–794. [Online]. Available: <https://dl.acm.org/doi/10.1145/2939672.2939785>
- [14] J. B. G. Brito, G. B. Bucco, R. Heldt, J. L. Becker, C. S. Silveira, F. B. Luce, and M. J. Anzanello, "A framework to improve churn prediction performance in retail banking," *Financial Innovation*, vol. 10, 2024. [Online]. Available: <https://link.springer.com/article/10.1186/s40854-023-00558-3>
- [15] A. Barsotti, G. Gianini, C. Mio, J. Lin, H. Babbar, A. Singh, F. Taher, and E. Damiani, "A decade of churn prediction techniques in the TelCo domain: A survey," *SN Computer Science*, vol. 5, 2024. [Online]. Available: <https://link.springer.com/article/10.1007/s42979-024-02722-7>
- [16] P. Lalwani, M. K. Mishra, J. S. Chadha, and P. Sethi, "Customer churn prediction system: a machine learning approach," *Computing*, vol. 104, no. 2, 2022. [Online]. Available: <https://link.springer.com/article/10.1007/s00607-021-00908-y>
- [17] T. Fawcett, "An introduction to ROC analysis," *Pattern Recognition Letters*, vol. 27, no. 8, 2006. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S016786550500303X>
- [18] A. G. Martín, A. Fernández-Isabel, I. Martín de Diego, and M. Beltrán, "A survey for user behavior analysis based on machine learning techniques: current models and applications," *Applied Intelligence*, vol. 51, 2021. [Online]. Available: <https://link.springer.com/article/10.1007/s10489-020-02160-x>
- [19] Z. Gharibshah and X. Zhu, "User response prediction in online advertising," *ACM Computing Surveys*, vol. 54, no. 3, 2022. [Online]. Available: <https://dl.acm.org/doi/10.1145/3446662>
- [20] M. Imani, M. Joudaki, A. Beikmohammadi, and H. R. Arabnia, "Customer churn prediction: A systematic review of recent advances, trends, and challenges in machine learning and deep learning," *Machine Learning and Knowledge Extraction*, vol. 7, no. 3, 2025. [Online]. Available: <https://www.mdpi.com/2504-4990/7/3/105>
- [21] L. D. Hollebeek, M. S. Glynn, and R. J. Brodie, "Consumer brand engagement in social media: Conceptualization, scale development and validation," *Journal of Interactive Marketing*, vol. 28, no. 2, 2014. [Online]. Available: <https://www.sciencedirect.com/science/article/abs/pii/S1094996813000649>
- [22] L. D. Hollebeek and K. Macky, "Digital content marketing's role in fostering consumer engagement, trust, and value: Framework, fundamental propositions, and implications," *Journal of Interactive Marketing*, vol. 45, 2019. [Online]. Available: https://www.researchgate.net/publication/330801358_Digital_Content_Marketing's_Role_in_Fostering_Consumer_Engagement_Trust_and_Value_Framework_Fundamental_Propositions_and_Implications
- [23] N. V. Chawla, K. W. Bowyer, L. O. Hall, and W. P. Kegelmeyer, "SMOTE: synthetic minority over-sampling technique," *Journal of Artificial Intelligence Research*, vol. 16, pp. 321–357, 2002. [Online]. Available: <https://www.jair.org/index.php/jair/article/view/10302>